

9.20 The expected complete-data log-likelihood function (9.62) is given by:

$$\begin{aligned}\mathbb{E}[\ln p(\mathbf{t}, \mathbf{w}|\alpha, \beta)] &= \frac{M}{2} \ln \left(\frac{\alpha}{2\pi} \right) - \frac{\alpha}{2} \mathbb{E}[\mathbf{w}^T \mathbf{w}] \\ &+ \frac{N}{2} \ln \left(\frac{\beta}{2\pi} \right) - \frac{\beta}{2} \sum_{n=1}^N \mathbb{E}[(t_n - \mathbf{w}^T \phi_n)^2]\end{aligned}$$

Taking derivative w.r.t α , the last 2 terms disappear, and we get:

$$\begin{aligned}\frac{\partial \mathbb{E}[\ln p(\mathbf{t}, \mathbf{w}|\alpha, \beta)]}{\partial \alpha} &= \frac{M}{2} \frac{\partial}{\partial \alpha} \left(\ln \left(\frac{\alpha}{2\pi} \right) \right) - \frac{\partial}{\partial \alpha} \left(\frac{\alpha}{2} \mathbb{E}[\mathbf{w}^T \mathbf{w}] \right) \\ &= \frac{M}{2} \cdot \frac{1}{\alpha/(2\pi)} \cdot \frac{\partial}{\partial \alpha} \left(\frac{\alpha}{2\pi} \right) - \frac{1}{2} \mathbb{E}[\mathbf{w}^T \mathbf{w}] \\ &= \frac{M}{2} \cdot \frac{2\pi}{\alpha} \cdot \frac{1}{2\pi} - \frac{\mathbb{E}[\mathbf{w}^T \mathbf{w}]}{2} \\ &= \frac{M}{2\alpha} - \frac{\mathbb{E}[\mathbf{w}^T \mathbf{w}]}{2}\end{aligned}$$

Setting this derivative to 0, we get:

$$\begin{aligned}\frac{M}{2\alpha} - \frac{\mathbb{E}[\mathbf{w}^T \mathbf{w}]}{2} &= 0 \\ \implies \frac{M}{\alpha} &= \mathbb{E}[\mathbf{w}^T \mathbf{w}] \\ \implies \alpha &= \frac{M}{\mathbb{E}[\mathbf{w}^T \mathbf{w}]}\end{aligned}$$

From identity (378) of The Matrix Cookbook, we know that assuming $\mathbf{x} \sim \mathcal{N}(\mathbf{m}, \mathbf{\Sigma})$,

$$\mathbb{E}[\mathbf{x}^T \mathbf{A} \mathbf{x}] = \text{Tr}(\mathbf{A} \mathbf{\Sigma}) + \mathbf{m}^T \mathbf{A} \mathbf{m}$$

If we let $\mathbf{A} = \mathbf{I}$, and apply this identity to $\mathbf{w} \sim \mathcal{N}(\mathbf{m}_N, \mathbf{S}_N)$, we get:

$$\mathbb{E}[\mathbf{w}^T \mathbf{w}] = \text{Tr}(\mathbf{S}_N) + \mathbf{m}_N^T \mathbf{m}_N$$

Substituting this back into the result for α , we get:

$$\alpha = \frac{M}{Tr(\mathbf{S}_N) + \mathbf{m}_N^T \mathbf{m}_N}$$

which is the same as (9.63).

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